

# AN1358 (AN6562), AN1358S (AN6562S)

## Dual Operational Amplifiers

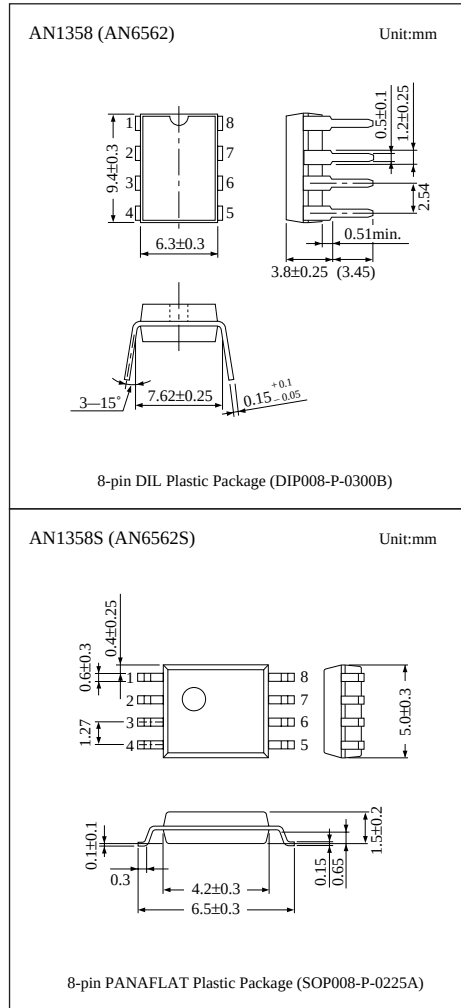
### ■ Overview

The AN1358 (AN6562) and AN1358S (AN6562S) are dual operational amplifiers with two phase compensation circuits built-in, have a wide range of operating supply voltage, and can operate on a single power supply.

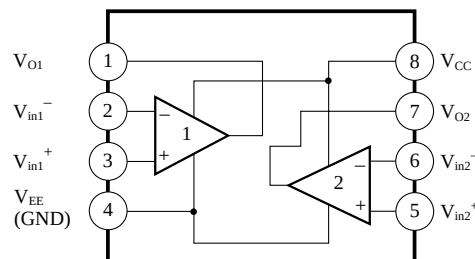
They have electrical characteristics equivalent to those of the conventional operational amplifiers, and are low-powered and suitable for application to various circuits. Note) The type numbers in ( ) are old ones.

### ■ Features

- Built-in phase compensation circuits
- Wide range of input voltage: 0V to  $V_{CC}-1.5V$
- Wide range of operating supply voltage:  
Single power supply: 3 to 30V  
Dual power supply:  $\pm 1.5$  to  $\pm 15V$



### ■ Block Diagram



## ■ Absolute Maximum Ratings (Ta=25°C)

| Parameter                     |                   | Symbol    | Rating       | Unit |
|-------------------------------|-------------------|-----------|--------------|------|
| Supply voltage                |                   | $V_{CC}$  | 32           | V    |
| Differential input voltage    |                   | $V_{ID}$  | 32           | V    |
| Common-mode input voltage     |                   | $V_{ICM}$ | − 0.3 to +32 | V    |
| Output voltage                |                   | $V_O$     | 24           | V    |
| Power dissipation             | AN1358 (AN6562)   | $P_D$     | 350          | mW   |
|                               | AN1358 (AN6562S)  |           | 360          |      |
| Operating ambient temperature |                   | $T_{opr}$ | −20 to +75   | °C   |
| Storage temperature           | AN1358 (AN6562)   | $T_{stg}$ | −55 to +150  | °C   |
|                               | AN1358S (AN6562S) |           | −55 to +125  |      |

## ■ Recommended Operating Range (Ta=25°C)

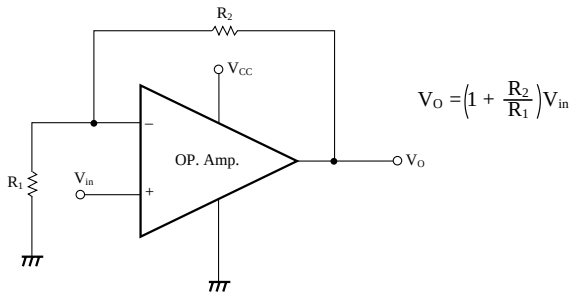
| Parameter                      | Symbol   | Range                           |
|--------------------------------|----------|---------------------------------|
| Operating supply voltage range | $V_{CC}$ | Single power supply 3V to 30V   |
|                                |          | Dual power supply ±1.5V to ±15V |

## ■ Electrical Characteristics (V<sub>CC</sub>=5V, Ta=25°C)

| Parameter                       | Symbol          | Condition                  | min          | typ | max          | Unit |
|---------------------------------|-----------------|----------------------------|--------------|-----|--------------|------|
| Input offset voltage            | $V_{I(offset)}$ | $R_S=50\Omega$             | —            | 2   | 7            | mV   |
| Input bias current              | $I_{bias}$      |                            | —            | —   | 250          | nA   |
| Input offset current            | $I_{IO}$        |                            | —            | —   | 50           | nA   |
| Common-mode input voltage width | $V_{CM}$        |                            | 0            | —   | $V_{CC}-1.5$ | V    |
| Supply current                  | $I_{CC}$        | $R_L=\infty$               | —            | 0.6 | 1.2          | mA   |
| Voltage gain                    | $G_V$           | $R_L \geq 2k\Omega$        | —            | 100 | —            | dB   |
| Maximum output voltage          | $V_{O(max)}$    | $R_L \geq 2k\Omega$        | $V_{CC}-1.5$ | —   | —            | V    |
| Common-mode rejection ratio     | CMR             |                            | 65           | 85  | —            | dB   |
| Supply voltage rejection ratio  | SVR             |                            | 65           | 100 | —            | dB   |
| Channel separation              | CS              | f=1 to 20kHz               | —            | 120 | —            | dB   |
| Output source current           | $I_{O(source)}$ | $V_{in}^+=1V, V_{in}^-=0V$ | 20           | 40  | —            | mA   |
| Output sink current             | $I_{SINK}$      | $V_{in}^+=0V, V_{in}^-=1V$ | 10           | 20  | —            | mA   |

## ■ Application Circuit

Non-inverting Amplifier



## ■ Pin Descriptions

| Pin No. | Pin name                      |
|---------|-------------------------------|
| 1       | Ch.1 output pin               |
| 2       | Ch.1 inverting input pin      |
| 3       | Ch.1 non-inverting input pin  |
| 4       | Negative supply voltage (GND) |
| 5       | Ch.2 non-inverting input pin  |
| 6       | Ch.2 inverting input pin      |
| 7       | Ch.2 output pin               |
| 8       | Positive supply voltage       |

## ■ Characteristics Curve

